

|         |                                                                                                                        |
|---------|------------------------------------------------------------------------------------------------------------------------|
| Ex.no:8 | <h1 style="text-align: center;">MINI PROJECT – HOTEL BOOKINGS</h1> <h2 style="text-align: center;">DATA ANALYTICS</h2> |
|         |                                                                                                                        |

### Domain and Problem Statement:

**Domain:** Hospitality Analytics & Tourism Management

**Problem Statement:** Hotel chains face revenue instability caused by high cancellation rates, fluctuating seasonal demand, and unoptimized pricing strategies (ADR). This project addresses these challenges by transforming raw booking transactional records into structured business intelligence across Power BI and Tableau to evaluate booking performance, geographic trends, and customer segments.

### Dataset Description:

**Dataset Name:** Hotel Booking Demand Dataset.

**Granularity:** Each row represents a single hotel reservation transaction.

**Timeframe Covered:** 2015 to 2017.

#### Key Entities & Attributes:

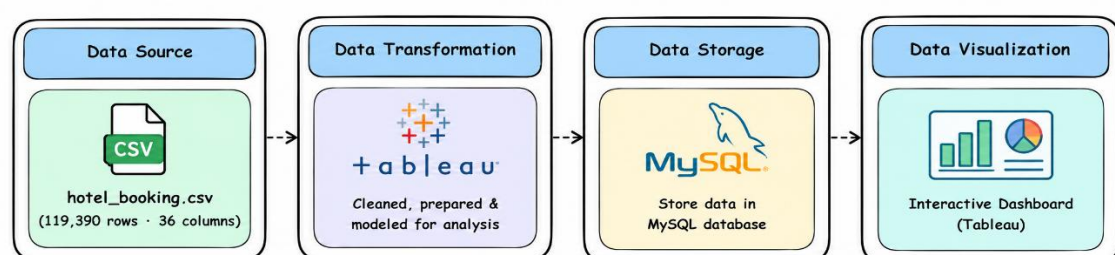
- **Demographics:** country, customer\_type, market\_segment, distribution\_channel.
- **Booking Details:** hotel, is\_canceled, lead\_time, arrival\_date\_year, arrival\_date\_month, arrival\_date\_day\_of\_month.
- **Financials:** adr (Average Daily Rate), deposit\_type, stays\_in\_weekend\_nights, stays\_in\_week\_nights.

### Data Source:

**Primary Source:** Kaggle Open Datasets (Hotel Booking Demand).

**Storage Infrastructure:** Cleaned transactional records hosted within a local MySQL database instance (hotel\_analytics).

### Architecture:



## **Program:**

### **1. Python – Data Cleaning & Transformation**

```
import pandas as pd

from sqlalchemy import create_engine, text

# Load dataset
df = pd.read_csv("hotel_bookings.csv")

# Create a copy for processing
hotel = df.copy()

# Remove unnecessary columns
hotel.drop(columns=["agent", "company"], inplace=True)

# Handle missing values
hotel["children"] = hotel["children"].fillna(0)
hotel["country"] = hotel["country"].fillna("Unknown")

# Remove duplicate records
hotel.drop_duplicates(inplace=True)

# Create arrival date
hotel["arrival_date"] = pd.to_datetime(
    hotel["arrival_date_year"].astype(str) + "-" +
    hotel["arrival_date_month"] + "-" +
    hotel["arrival_date_day_of_month"].astype(str),
    format="%Y-%B-%d"
)

# Create derived columns
hotel["total_nights"] = (
    hotel["stays_in_weekend_nights"] +
    hotel["stays_in_week_nights"]
)
```

```
hotel["total_guests"] = (  
    hotel["adults"] +  
    hotel["children"] +  
    hotel["babies"]  
)
```

```
hotel["total_extra_requests"] = (  
    hotel["total_of_special_requests"] +  
    hotel["required_car_parking_spaces"]  
)
```

```
# Standardize column names
```

```
hotel.columns = (  
    hotel.columns  
    .str.lower()  
    .str.strip()  
    .str.replace(" ", "_")  
)
```

```
print("Original Shape:", df.shape)
```

```
print("Cleaned Shape:", hotel.shape)
```

```
hotel.head()
```

## **2. MySQL – Database and Data Loading**

### **MySQL Command Prompt:**

```
net start MySQL80
```

```
mysql -u root -p
```

### **MySQL:**

```
CREATE DATABASE IF NOT EXISTS hotel_analytics;
```

### **Python/Jupyter:**

```
username = "root"
```

```
password = "YOUR_MYSQL_PASSWORD"
```

```
host = "localhost"
```

```
port = 3306
```

```
# Connect to MySQL database
```

```
db_engine = create_engine(
```

```
    f'mysql+pymysql://{username}:{password}@{host}:{port}/hotel_analytics"
```

```
)
```

```
# Load cleaned data into MySQL
```

```
hotel.to_sql(
```

```
    "hotel_bookings",
```

```
    con=db_engine,
```

```
    if_exists="replace",
```

```
    index=False,
```

```
    chunksize=5000
```

```
)
```

```
print("Data loaded successfully into MySQL.")
```

### **Verify:**

```
pd.read_sql(
```

```
    "SELECT COUNT(*) AS total_rows FROM hotel_bookings",
```

```
    db_engine
```

```
)
```

## Procedure:

### Step 1: Data Preparation & Preprocessing (Python / Pandas)

- Load the raw Kaggle dataset into a Jupyter Notebook / Python script using `pandas.read_csv()`.
- Inspect for null values and handle missing data in country, children, and agent columns.
- Feature engineer total stay length by combining weekend and weeknight stays:

$$\text{total\_stays} = \text{stays\_in\_weekend\_nights} + \text{stays\_in\_week\_nights}$$

- Filter out invalid rows (e.g., zero total guests or negative ADR entries) and export the cleaned DataFrame to MySQL using sqlalchemy and `to_sql()`.

```
Rows: 119390
Columns: 36

[3]:
```

|   | hotel        | is_canceled | lead_time | arrival_date_year | arrival_date_month | arrival_date_week_number | arrival_date_day_of_month | stays_in_weekend_nights | stays_in_v |
|---|--------------|-------------|-----------|-------------------|--------------------|--------------------------|---------------------------|-------------------------|------------|
| 0 | Resort Hotel | 0           | 342       | 2015              | July               | 27                       | 1                         | 0                       |            |
| 1 | Resort Hotel | 0           | 737       | 2015              | July               | 27                       | 1                         | 0                       |            |
| 2 | Resort Hotel | 0           | 7         | 2015              | July               | 27                       | 1                         | 0                       |            |
| 3 | Resort Hotel | 0           | 13        | 2015              | July               | 27                       | 1                         | 0                       |            |
| 4 | Resort Hotel | 0           | 14        | 2015              | July               | 27                       | 1                         | 0                       |            |

5 rows × 36 columns

```
Duplicate records before cleaning: 32020
Duplicate records after cleaning: 0
Original rows: 119390
Cleaned rows: 87370
Columns: 34

Missing values:
Series([], dtype: int64)

i]:
```

|   | hotel        | is_canceled | lead_time | arrival_date_year | arrival_date_month | arrival_date_week_number | arrival_date_day_of_month | stays_in_weekend_nights | stays_in_v |
|---|--------------|-------------|-----------|-------------------|--------------------|--------------------------|---------------------------|-------------------------|------------|
| 0 | Resort Hotel | 0           | 342       | 2015              | July               | 27                       | 1                         | 0                       |            |
| 1 | Resort Hotel | 0           | 737       | 2015              | July               | 27                       | 1                         | 0                       |            |
| 2 | Resort Hotel | 0           | 7         | 2015              | July               | 27                       | 1                         | 0                       |            |
| 3 | Resort Hotel | 0           | 13        | 2015              | July               | 27                       | 1                         | 0                       |            |
| 4 | Resort Hotel | 0           | 14        | 2015              | July               | 27                       | 1                         | 0                       |            |

5 rows × 34 columns



## Step 2: SQL Database Connection & Data Ingestion

1. Open Power BI Desktop → Click **Get Data** → Select **MySQL Database**.
2. Enter the **Server** (localhost:3306) and **Database Name** (hotel\_analytics).
3. Enter authentication credentials, select the cleaned flat table (hotel\_bookings / hotel\_analytics), and click **Transform Data** to open Power Query

### MySQL database

Server

localhost:3306

Database

hotel\_analytics

Advanced options

OK

Cancel

### Navigator

Display Options

localhost:3306: hotel\_analytics [1]

☒ hotel\_analytics.hotel\_bookings

#### hotel\_analytics.hotel\_bookings

| hotel        | is_canceled | lead_time | arrival_date_year | arrival_date_mo |
|--------------|-------------|-----------|-------------------|-----------------|
| Resort Hotel | 0           | 342       | 2015              | July            |
| Resort Hotel | 0           | 737       | 2015              | July            |
| Resort Hotel | 0           | 7         | 2015              | July            |
| Resort Hotel | 0           | 13        | 2015              | July            |
| Resort Hotel | 0           | 14        | 2015              | July            |
| Resort Hotel | 0           | 0         | 2015              | July            |
| Resort Hotel | 0           | 9         | 2015              | July            |
| Resort Hotel | 1           | 85        | 2015              | July            |
| Resort Hotel | 1           | 75        | 2015              | July            |
| Resort Hotel | 1           | 23        | 2015              | July            |
| Resort Hotel | 0           | 35        | 2015              | July            |
| Resort Hotel | 0           | 68        | 2015              | July            |
| Resort Hotel | 0           | 18        | 2015              | July            |
| Resort Hotel | 0           | 37        | 2015              | July            |

The data in the preview has been truncated due to size limits.

Select Related Tables

Load

Transform Data

Cancel

Table: hotel\_analytics.hotel\_bookings (107,319 rows)

| hotel        | is_canceled | lead_time | arrival_date_year | arrival_date_month | arrival_date_week_number | arrival_date_day_of_month | days_in_weekend_stay | days_in_week_stay | adults | children | babies | meal |
|--------------|-------------|-----------|-------------------|--------------------|--------------------------|---------------------------|----------------------|-------------------|--------|----------|--------|------|
| Resort Hotel | 0           | 72        | 2017              | August             | 31                       | 31                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 0           | 0         | 2017              | October            | 44                       | 31                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 0           | 0         | 2017              | October            | 44                       | 30                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 15        | 2016              | January            | 1                        | 31                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 50        | 2016              | March              | 10                       | 3                         | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 18        | 2016              | March              | 12                       | 16                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 42        | 2016              | March              | 16                       | 30                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 46        | 2016              | April              | 10                       | 9                         | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 216       | 2016              | May                | 22                       | 26                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 217       | 2016              | May                | 23                       | 27                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 0         | 2016              | June               | 26                       | 25                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 61        | 2016              | September          | 26                       | 1                         | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 70        | 2017              | January            | 4                        | 30                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 22        | 2017              | January            | 4                        | 30                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 26        | 2017              | February           | 9                        | 11                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 22        | 2017              | February           | 9                        | 11                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 47        | 2017              | February           | 7                        | 14                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 23        | 2017              | February           | 7                        | 12                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 26        | 2017              | February           | 7                        | 16                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 60        | 2017              | April              | 10                       | 3                         | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 11        | 2017              | May                | 21                       | 22                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 32        | 2017              | May                | 21                       | 24                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 7         | 2017              | June               | 27                       | 2                         | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 0         | 2017              | June               | 23                       | 0                         | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 368       | 2017              | June               | 24                       | 17                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 46        | 2017              | August             | 23                       | 19                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 1           | 208       | 2017              | August             | 20                       | 19                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 0           | 0         | 2016              | August             | 23                       | 12                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 0           | 0         | 2016              | September          | 26                       | 16                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 0           | 1         | 2016              | October            | 44                       | 30                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 0           | 1         | 2016              | November           | 47                       | 11                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |
| Resort Hotel | 0           | 0         | 2016              | November           | 48                       | 25                        | 0                    | 0                 | 1      | 0        | 0      | 0.00 |

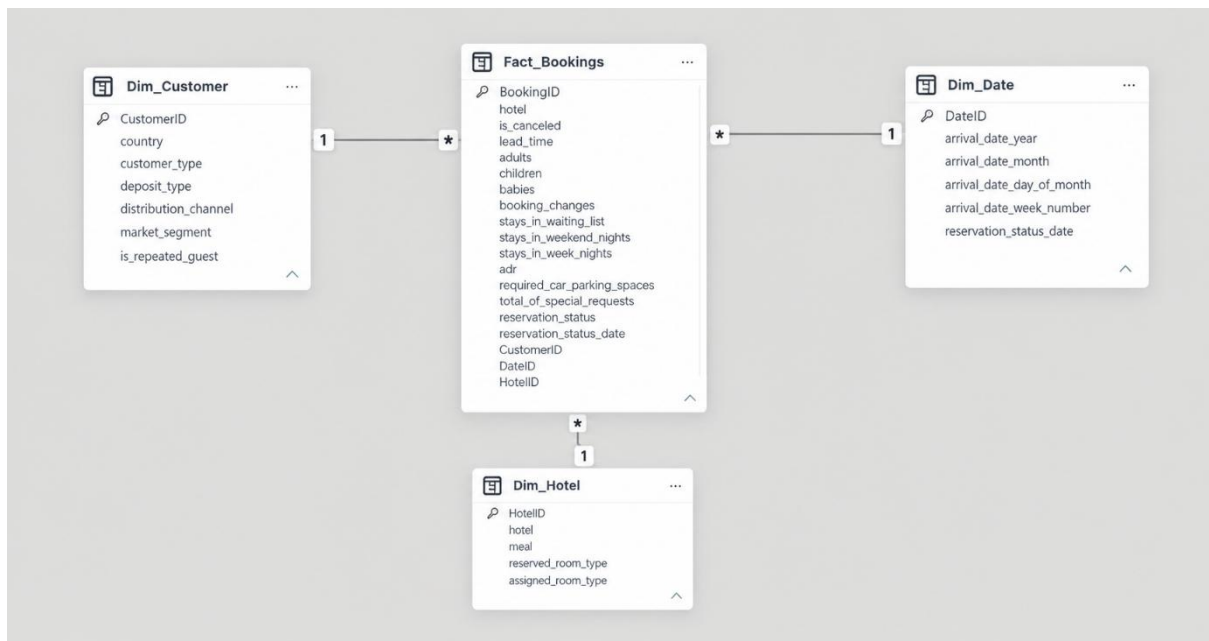
### Step 3: Star Schema Data Modeling (Power BI)

#### 1. Table Splitting in Power Query:

- Duplicate the imported flat table (hotel\_bookings) three times to create separate entities: Fact\_Bookings, Dim\_Customer, Dim\_Hotel, and Dim\_Date.
- In Dim\_Customer, keep customer\_type, market\_segment, distribution\_channel, and country. Remove duplicates and add an Index Column as CustomerID.
- In Dim\_Hotel, keep hotel. Remove duplicates and add an Index Column as HotelID.
- In Fact\_Bookings, merge with dimension tables to map CustomerID and HotelID, then remove raw text columns.
- Click **Close & Apply**.

#### 2. Establishing Relationships in Model View:

- Open the **Model View** tab in Power BI.
- Connect Dim\_Customer[CustomerID] → Fact\_Bookings[CustomerID] (1:\*).
- Connect Dim\_Hotel[HotelID] → Fact\_Bookings[HotelID] (1:\*).
- Connect Dim\_Date[Date] → Fact\_Bookings[arrival\_date] (1:\*).
- Hide foreign key columns inside Fact\_Bookings to keep the report fields pane organized.



## Step 4: Core Calculations & Measures (DAX)

### Power BI Measures (DAX)

Create a dedicated \_Measures table and add the following DAX calculations:

- **Total Bookings:**

```
1 Total Bookings = COUNTROWS(hotel_booking)
```

- **Total Revenue:**

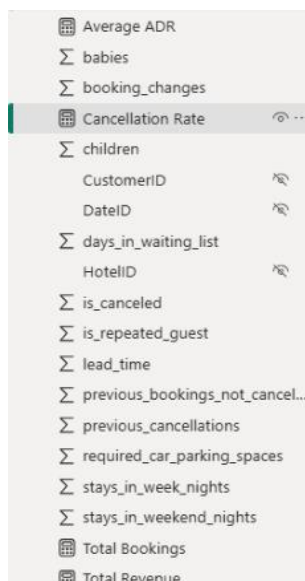
```
1 Total Revenue =  
2 SUMX(  
3     hotel_booking,  
4     (hotel_booking[stays_in_weekend_nights] +  
5     hotel_booking[stays_in_week_nights]) *  
6     hotel_booking[adr]  
7 )
```

- **Cancellation Rate (%):**

```
1 Cancellation Rate =  
2 DIVIDE(  
3     CALCULATE(  
4         COUNTROWS(hotel_booking),  
5         hotel_booking[is_canceled] = 1  
6     ),  
7     [Total Bookings],  
8     0  
9 )
```

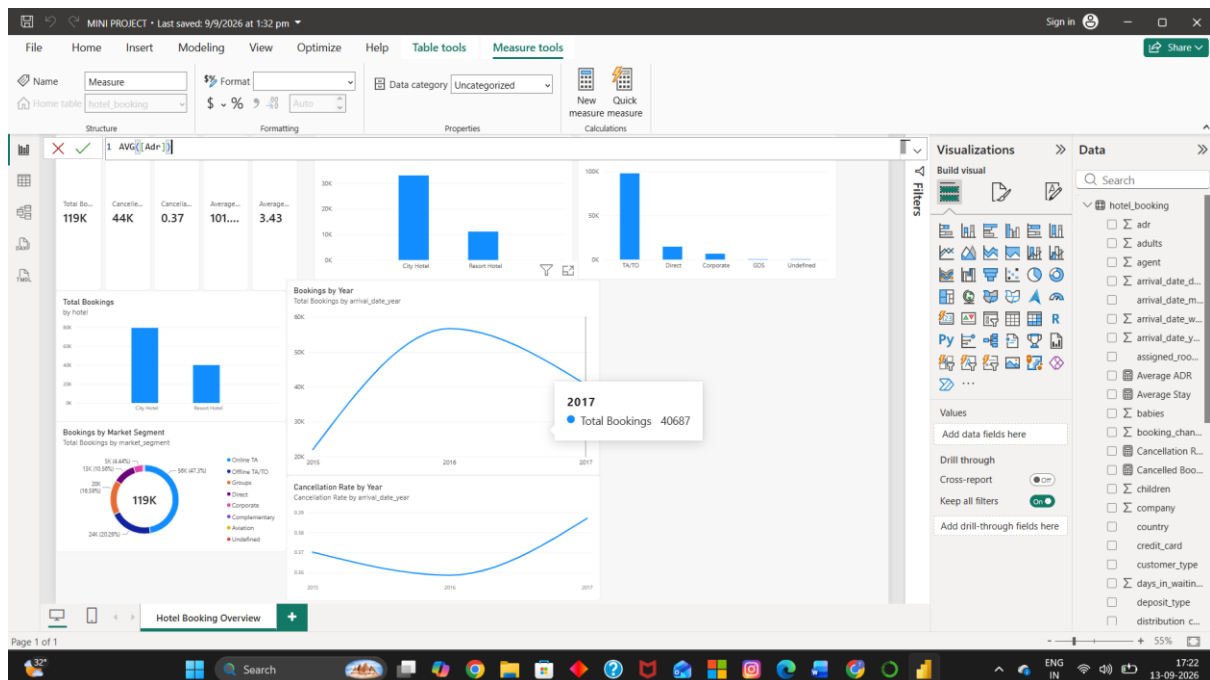
- **Average Daily Rate (ADR):**

```
1 AVG([Adr])
```



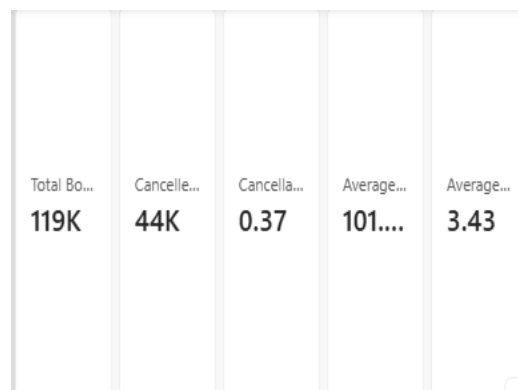
|                                   |
|-----------------------------------|
| Average ADR                       |
| Σ babies                          |
| Σ booking_changes                 |
| Cancellation Rate                 |
| Σ children                        |
| CustomerID                        |
| DateID                            |
| Σ days_in_waiting_list            |
| HotelID                           |
| Σ is_canceled                     |
| Σ is_repeated_guest               |
| Σ lead_time                       |
| Σ previous_bookings_not_cancel... |
| Σ previous_cancellations          |
| Σ required_car_parking_spaces     |
| Σ stays_in_week_nights            |
| Σ stays_in_weekend_nights         |
| Total Bookings                    |
| Total Revenue                     |

## Step 5: Power BI Final Executive Dashboard & Drill-Through

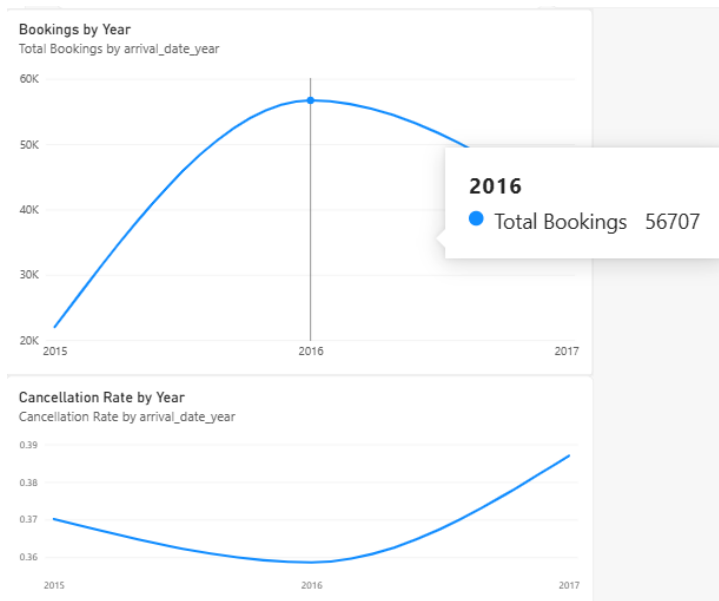


### 1. Executive Canvas:

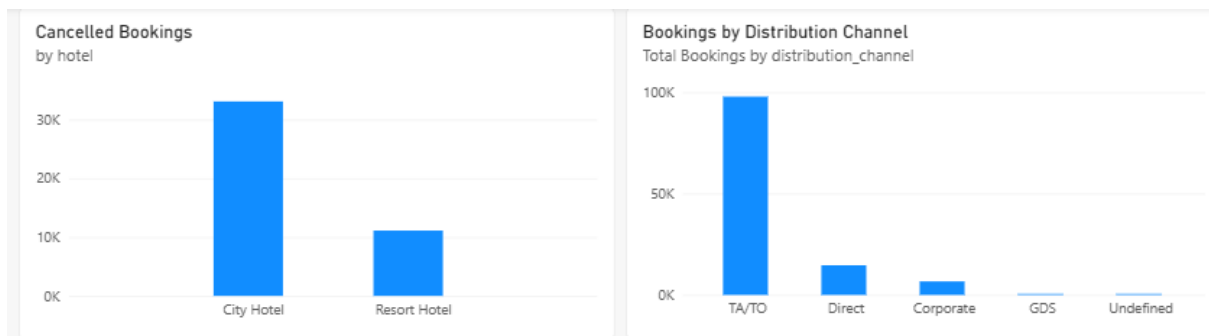
- Place **Card Visuals** across the top: Total Revenue, Total Bookings, Cancellation Rate, Average ADR.



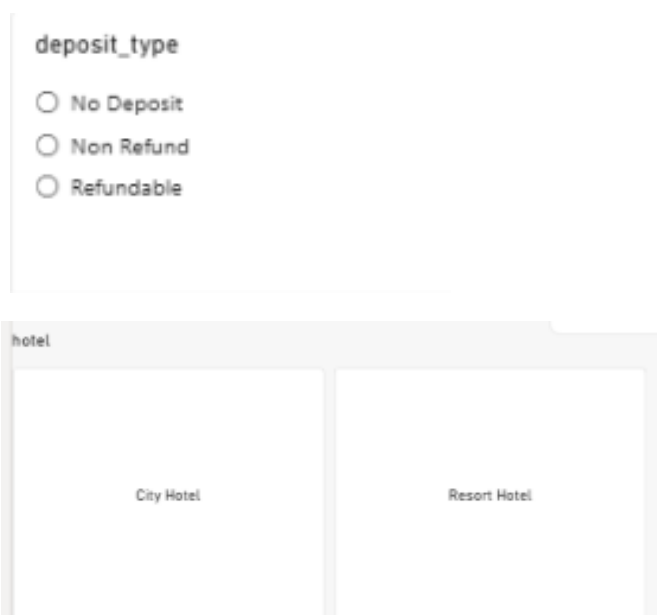
- Add a **Line Chart** mapped to Dim\_Date[arrival\_date] with hierarchy drill-down enabled (Year → Quarter → Month).



- Add a **Clustered Bar Chart** for Cancellation Rate by Dim\_Hotel[hotel].

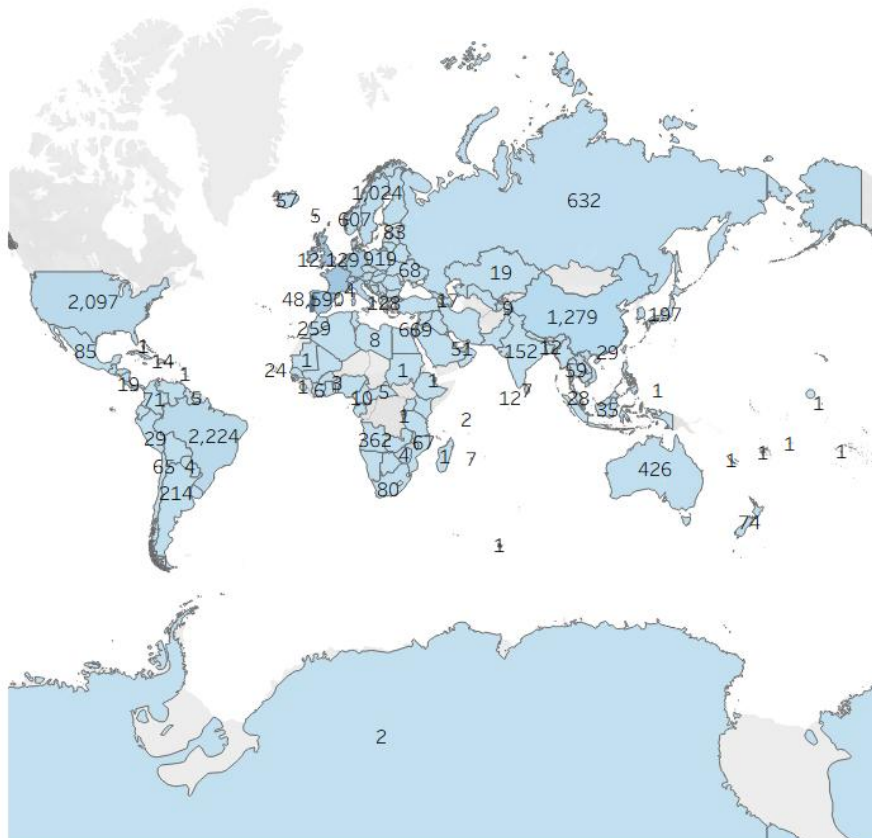
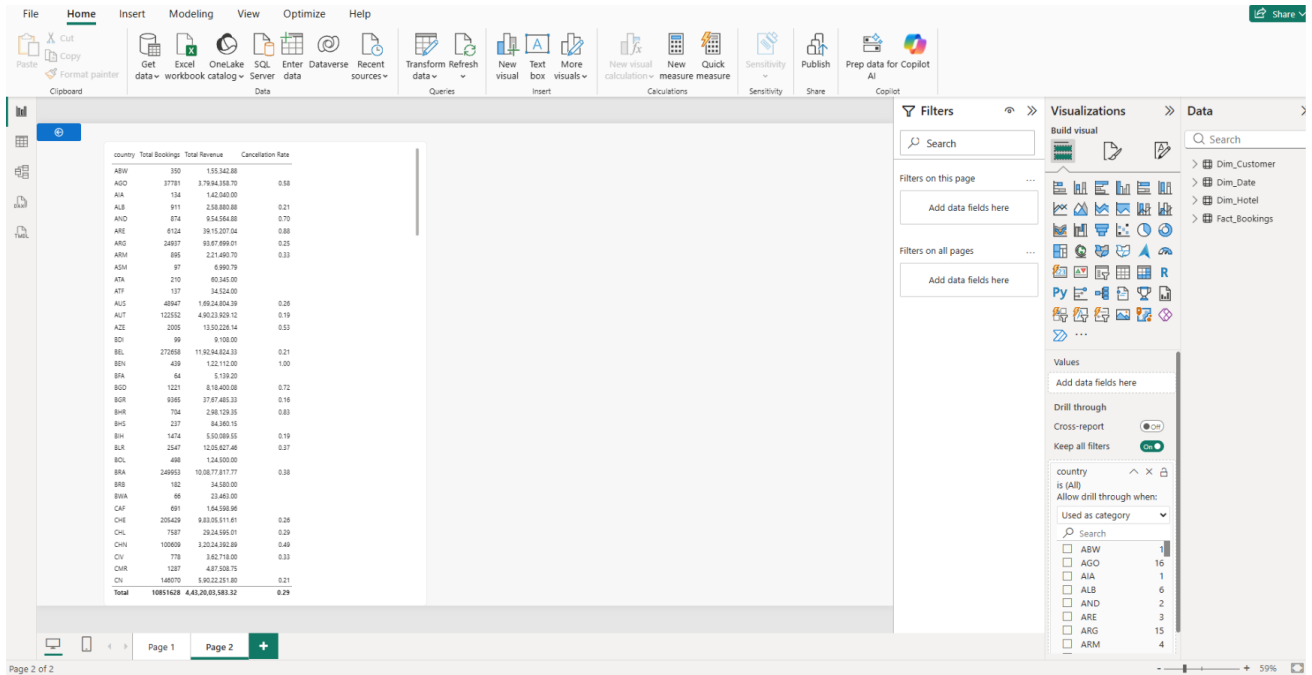


- Add **Slicers** for hotel, deposit\_type



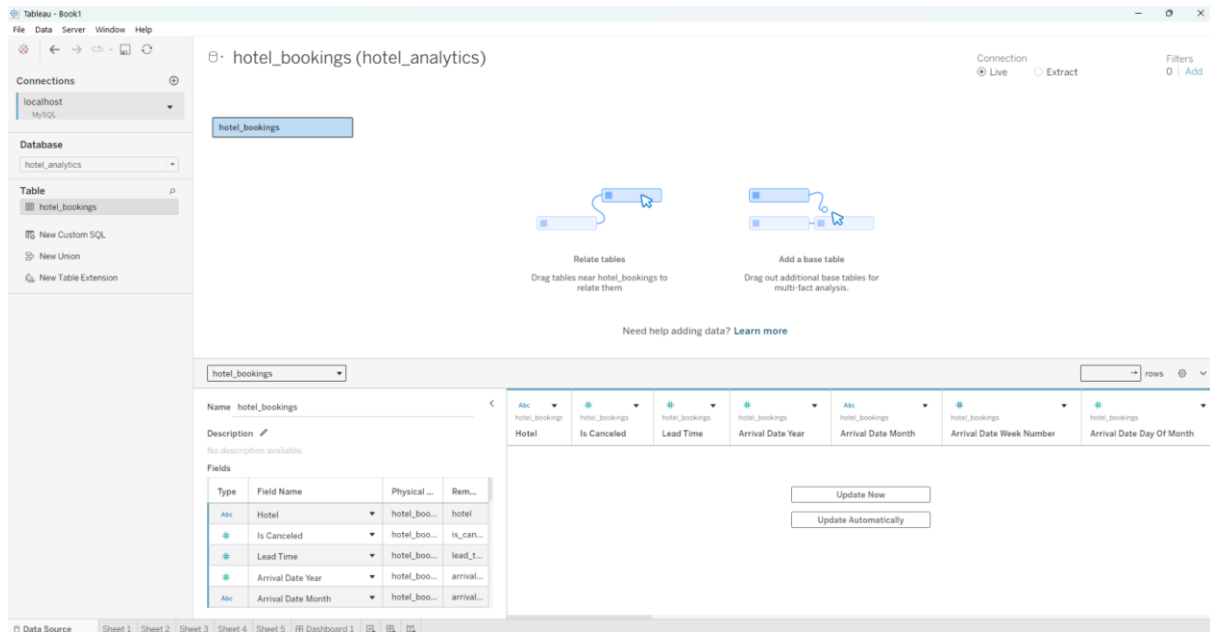
## 2. Country Details Drill-Through Page:

- Create a second page named Country Details.
- Drag Dim\_Customer[country] into the **Drill-through** field bucket.
- Add a detailed Matrix/Table visual showing country, Total Bookings, Total Revenue, and Cancellation Rate.

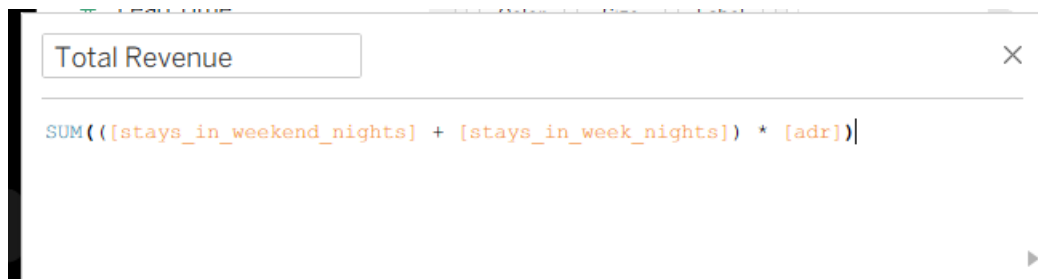


## Step 6: Tableau Data Connection & Preparation

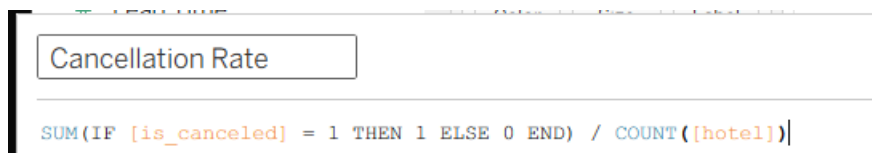
1. Open Tableau Desktop / Tableau Public.
2. Under **To a Server**, select **MySQL**
3. Enter Server (localhost), Database (hotel\_analytics), and user credentials to pull the cleaned dataset onto the canvas.



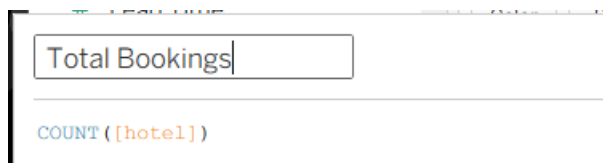
4. Go to any sheet and build calculated fields:
  - **Total Revenue**



- **Cancellation Rate**



- **Total Bookings**



○ **Average ADR**

Average ADR

×

AVG([Adr])|

- **Select Metric Parameter:** String list containing "Total Revenue", "Total Bookings", "Cancellation Rate".

Edit Parameter [Select Metric] 

×

Name

Select Metric

Properties

Data type

String

▼

Display format

Total Revenue

▼

Current value

Total Revenue

▼

Value when workbook opens

Current value

▼

Allowable values

☐ All

☒ List

☐ Range

| Value             | Display As        |  |
|-------------------|-------------------|--|
| Total Revenue     | Total Revenue     |  |
| Total Bookings    | Total Bookings    |  |
| Cancellation Rate | Cancellation Rate |  |
| Click to add      |                   |  |
|                   |                   |  |

☒ Fixed

☐ When workbook opens

Add values from ▼

Remove Selected

Cancel

OK

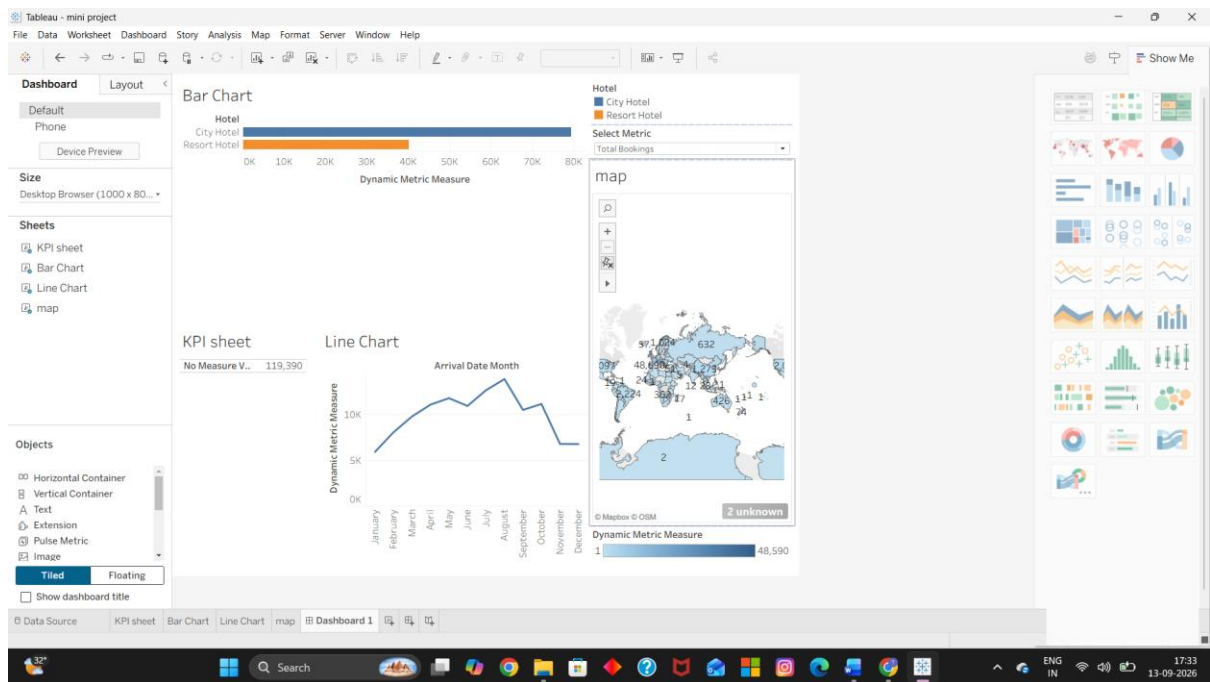
- **Dynamic Measure Formula:**

Dynamic Metric Measure

×

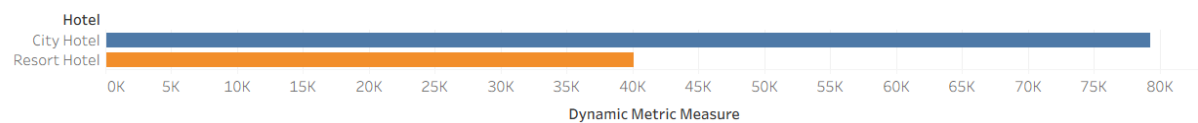
CASE [Select Metric]  
WHEN 'Total Revenue' THEN [Total Revenue]  
WHEN 'Total Bookings' THEN [Total Bookings]  
WHEN 'Cancellation Rate' THEN [Cancellation Rate]  
END|

## Step 7: Tableau Companion Dashboard & Interactive Story



1. **Worksheets:** Build **KPI Cards**, **Revenue Trend Line Chart** (using Dynamic Measure), **Hotel Cancellation Bar Chart**, and **Country Map**.

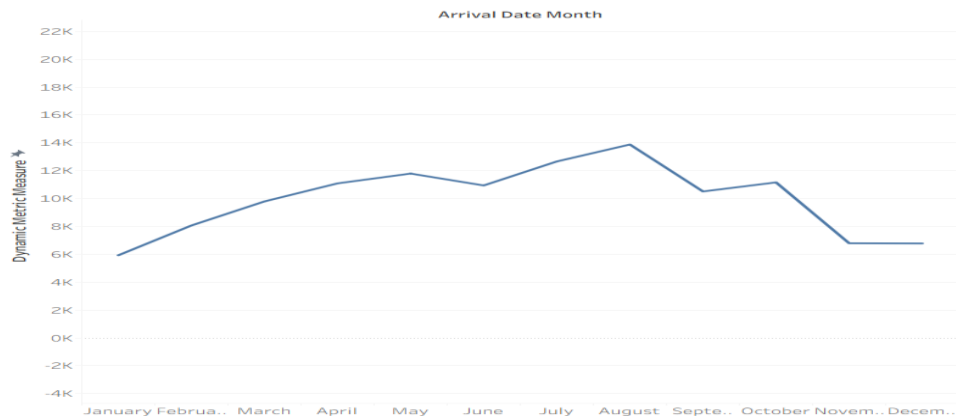
Bar Chart



KPI sheet

No Measure V.. 119,390

Line Chart



## 2. Global Filters & Actions:

- Drag Hotel, Deposit Type, and Arrival Date to Filters → set to **Apply to Worksheets** → **All Using This Data Source**.
- Drag sheets onto a new Dashboard tab and click the **Use as Filter** funnel icon on the Country Map visual.

Filter [Measure Names]

General

Enter search text

☐ Adr

☐ Adults

☒ Average ADR

☐ Babies

☐ Booking Changes

☒ Cancellation Rate

☐ Children

☐ Count of hotel\_bookings

☐ Days In Waiting List

☐ Dynamic Metric Measure

☐ Is Canceled

All

None

Summary

Field: [Measure Names]

Selection: Selected 4 of 24 values

Wildcard: All

Condition: None

Limit: None

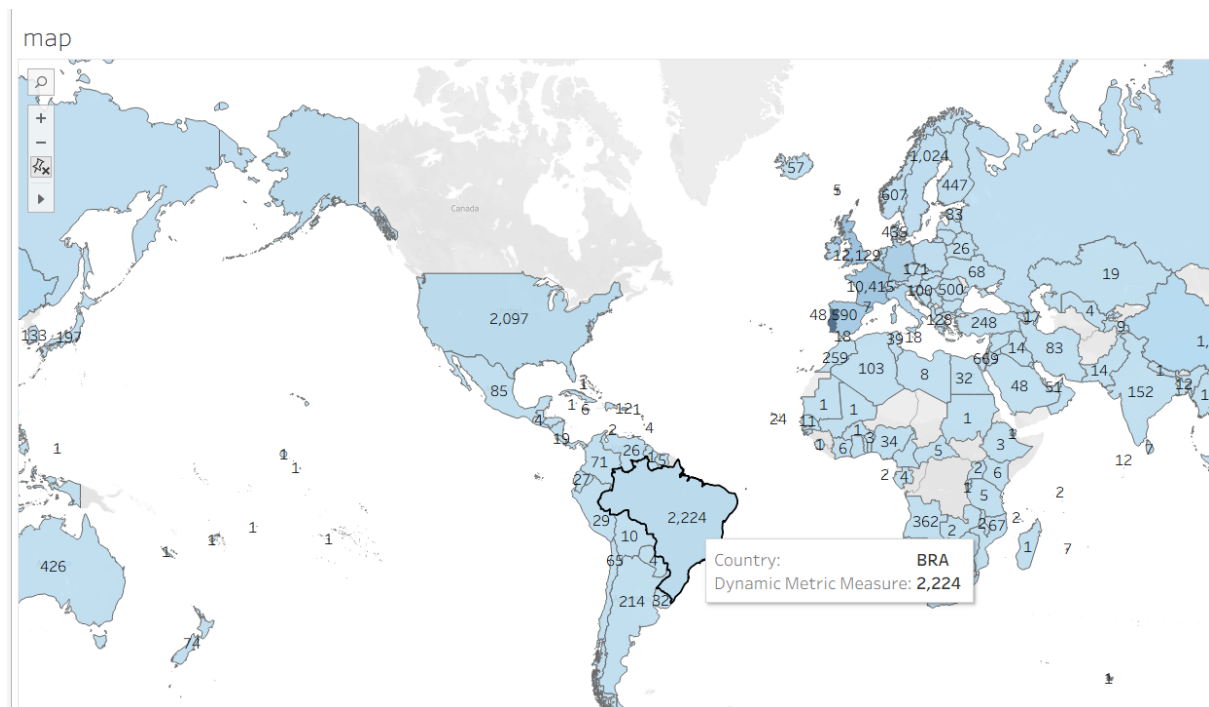
Reset

OK

Cancel

Apply

### 3. Drill-down map:



## Observed Insights:

- **Portfolio Revenue & Booking Volume:** The dataset captures **119.4K Total Bookings** generating **\$484.42M in Gross Revenue**, maintaining a portfolio **Average Daily Rate (ADR)** of **\$99.66**.
- **Cancellation Rates:** The overall portfolio reflects a **26% Cancellation Rate**, with **City Hotels** exhibiting higher cancellation volatility compared to **Resort Hotels**.
- **Seasonality Trends:** Booking volume and revenue peak significantly during the European summer window (**June–August**), followed by a sharp drop in winter (**December–January**).
- **Geographic Market Distribution:** Demand is heavily concentrated in Western Europe, with **Portugal (PRT)** serving as the top origin market, followed by **Great Britain (GBR)**, **France (FRA)**, and **Spain (ESP)**.